



Thesis Paper on the

"What is the impact of dark patterns in digital pre-purchase environments on consumer purchasing decisions?"

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Table of Contents

<i>Abstract</i>	1
 <i>Chapter 1: Introduction</i>	2
<i>1.1 Background and Context</i>	2
<i>1.2 Research Problem</i>	3
<i>1.3 Research Objectives</i>	3
<i>1.4 Research Question</i>	4
<i>1.5 Methodology Overview</i>	4
<i>1.6 Significance of the Study</i>	4
 <i>Chapter 2: Literature Review</i>	5
<i>2.1 Client Journey in E-Commerce</i>	5
2.1.1 Introduction	5
2.1.2 Awareness and Information Search	5
2.1.3 Evaluation of Alternatives	6
2.1.4 Purchase Decision and Post-Purchase Behavior	6
<i>2.2 Aspects of Dark Patterns and Impulsive Purchases</i>	7
2.2.1 Introduction	7
2.2.2 Types of Dark Patterns in E-Commerce	7
2.2.3 Psychological Mechanisms Behind Impulse Buying	8
2.2.4 Ethical and Regulatory Concerns	8

2.3 Negative Feedback and Sources of Disappointment in E-Commerce	9
2.3.1 Introduction	9
2.3.2 Common Complaints in Online Shopping	9
2.3.3 How Dark Patterns Affect Consumer Trust	10
2.3.4 Effectively Mitigating Consumer Disappointment	10
 Chapter 3: Overview	11
3.1 Introduction	11
3.2 Industry and Market Context	11
3.2.1 Growth of Emerging Advertising Trends	11
3.2.2 Rise of Dark Patterns in Digital Advertising	12
3.3 Theoretical and Conceptual Framework	13
3.3.1 Psychological Foundations of Dark Patterns	13
3.3.2 Ethical and Regulatory Concerns	14
3.4 Challenges and Concerns in the Industry	15
3.4.1 Damage Done to Consumer Trust and Brand Reputation	15
3.4.2 Legal and Ethical Discussions	15
3.4.3 Consumer Awareness and Digital Literacy	16
3.5 Justification for Effectiveness and Contribution of Research	16
3.6 Conclusion	16
 Chapter 4: Methodology	17
4.1 Research Design	17

<i>4.2 Qualitative Phase</i>	17
4.2.1 Data Collection: Semi-Structured Interviews	17
4.2.2 Interview Guide	19
4.2.3 Data Analysis	21
<i>4.3 Quantitative Phase</i>	21
4.3.1 Data Collection: Survey	21
4.3.2 Data Analysis	22
<i>4.4 Justification of Methodology</i>	22
<i>4.5 Ethical Considerations</i>	23
<i>4.6 Conclusion</i>	23
 <i>Chapter 5: Data Analysis and Key Findings</i>	25
5.1 Introduction	25
5.2 Qualitative Data Analysis	26
5.3 Quantitative Descriptive Analysis	30
5.4 Correlation and Pattern Analysis	31
5.5 Regression and Hypothesis Testing	32
5.6 Logistic Regression Analysis	33
5.7 Triangulation and Integrated Insights	34
5.8 Summary of Major Findings	35
 <i>Chapter 6: Discussion and Recommendations</i>	38
6.1 Discussion of Findings	38

6.1.1 Reinforcing the Role of Urgency in Behavioural Manipulation	38
6.1.2 Hidden Costs and the Violation of Trust	39
6.1.3 Transparency as the Cornerstone of Digital Trust	40
6.1.4 The Emotional Fallout from Visual Manipulation and Fake Discounts	40
6.1.5 Adaptive Behaviours: Digital Literacy as a Protective Layer	41
6.2 Recommendations	42
6.2.1 For Brands and E-Commerce Platforms	42
6.2.2 For Consumers	43
6.2.3 For Policymakers and Regulatory Bodies	44
Chapter 7: Limitations and Future Research Directions	45
7.1 Limitations of the Study	45
7.1.1 Sample Size and Representation	45
7.1.2 Platform and Environment Specificity	45
7.1.3 Self-Reported Bias and Social Desirability	46
7.1.4 Complexity of Causal Inference	46
7.1.5 Limited Scope of Dark Pattern Types	46
7.2 Future Research Designs	47
7.2.1 Longitudinal and Experimental Designs	47
7.2.2 Inclusion of Biometric or Behavioural Analytics	47
7.2.3 Comparative Cross-Cultural Studies	47
7.2.4 Vulnerable Population Focus	47

7.2.5 Policy Impact Assessments	47
<i>Chapter 8: Conclusion</i>	49

Abstract

This research delves deep and investigates the psychological and behavioural effects of dark patterns in digital pre-purchase environments on the Indian online consumers, with a focus on how such manipulative design tactics also sometimes push the customers to their extreme to force them to make a purchasing decision. This research also looks into how such tactics can also play a role with regards to trust, and post-purchase perceptions as well. By employing a mixed-methods approach, the study integrates findings from six qualitative interviews and a structured quantitative survey of 73 respondents as well. The qualitative phase helped uncover four main themes which were: decision making driven by urgency, hidden costs, erosion of trust and adaptive consumer strategies. These themes helped in formulating and developing the quantitative instrument for a wider audience, which measured variables such as susceptibility to urgency triggers, experiences with fake discounts and hidden charges, and finally brand trustworthiness.

The data was analysed with the help of descriptive statistics, Pearson correlation, linear regression, ANOVA, and logistic regression. From its analysis, it was understood that cues from urgency significantly affected trust levels, while transparency had also emerged as the key determinant for generating long term consumer loyalty. Correlation and regression findings also revealed that multiple dark patterns often co-occur, thus amplifying the negative effects that are felt by the customers. The study finally contributes to the growing discourse on digital ethics and customer protection by offering insights that are based on evidence for brands, regulatory bodies, and users that are navigating the increasingly manipulative landscape of social commerce.

Chapter 1: Introduction

1.1 Background and Context

In an era which is driven by algorithm based advertising and hyper-targeted consumer engagement, the involvement of dark patterns are well and truly felt, yet goes unnoticed by many, especially for those who are new to the digital age. It has become an integral component of the digital engagement strategies which are used by many platforms and brands. Coined by Brignull (2010), dark patterns refer to user interface designs that are built to mislead, coerce or deceive users into actions that they wouldn't have taken in a usually normal frame of mind. These patterns exploit cognitive biases among customers in order to maximise short-term conversions, which are usually done at the expense of long-term trust and user autonomy. Social media platforms such as Instagram, Facebook, Youtube and many others also serve as a fertile ground for such tactics to breed, given their immersive design and behavioural targeting capabilities.

In the Indian digital landscape, which is characterised by rapid e-commerce growth and high smartphone penetration, consumers fall particularly vulnerable to such deceptive tactics. While much of the academic literature on such dark patterns have been focused on the Western context, there is a relative paucity of empirical studies which look into the effects that they have on Indian consumers, especially within the cultural framework of price sensitivity, emerging digital literacy, and platform reliance. This study will look to fill that gap and provide valuable insights.

1.2 Research Problem

The increasing use of emergency cues such as, “Only 2 left!”, hidden fees, fake discount structures, and complex subscription cancellations often represent a pressing issue characterised by ethical and commercial concern. These manipulations not only reduce consumer trust but also distort fair competition in the market. The psychological burden of constant manipulation, paired with a lack of regulatory oversight in India, also raises significant questions revolving the long-term impact of these, on consumer behavior, platform credibility, and digital well-being.

1.3 Research Objectives:

By the end of this research, the following are to be achieved:

- To identify the most commonly experienced dark patterns in advertising by Indian online customers.
- To look into and examine how these patterns influence the minds of the customers and their decision-making, particularly impulse buying, trust formation and post-purchase regret.
- To statistically validate the relationship between exposure to dark patterns and indicators of brand distrust, from the point of view of the consumers.
- To assess the role of digital literacy and adaptive consumer behaviour in mitigating the impact caused through such patterns.

1.4 Research Question

With all of this taken into consideration, the following primary research question was created:

“What is the impact of dark patterns in digital pre-purchase environments on consumer purchasing decisions?”

1.5 Methodology Overview

This research employs a mixed-methods design, starting off with the qualitative stage, through which we were able to tap into a larger sample with the help of creating a quantitative study. The qualitative phase consisted of conducting 6 semi-structured interviews to explore subjective experiences. These findings helped create the design of a quantitative survey which was administered to 73 respondents. The survey results are analysed using descriptive statistics, Pearson correlation, linear regression, ANOVA and logistic regression to identify patterns and also test for a hypothesis which was derived from the qualitative phase.

1.6 Significance of the Study

The findings of this study would hold relevance for multiple stakeholders. For brands and marketers, it offers a data-driven critique of how the manipulative practices that they employ could lead to unintended long-term consequences. As far as the regulators are concerned, it provides empirical evidence to support the formation of ethical digital advertising frameworks. Finally, for the consumers, it reveals common tactics and also highlights the value of digital literacy as a defense mechanism to make better-informed decisions.

Chapter 2: Literature Review

2.1 Client Journey in E-Commerce

2.1.1 Introduction

The e-commerce industry has transformed consumer behavior by reshaping pathways between buyers and businesses. A client's journey consists of several stages, including the awareness, consideration, decision-making, purchase, and post-purchase phases, absolutely based on different determinants. Numerous elaborate measures have been tried by scholars to define and contextualise the realm of consumer decisions regarding the digital marketplace.

2.1.2 Awareness and Information Search

Consumer awareness about e-commerce is mostly defined by digital marketing strategies, algorithm-driven advertisement, and user-generated content in e-commerce. Mangiaracina, Perego, and Brugnoli (2016) pointed out that e-commerce sites help guide users through the initial search phase by offering product descriptions, filtering mechanisms, and customer reviews. Of these aspects, trust is crucial to making informed decisions in online transactions (Mangiaracina et al., 2016).

Recently, Hollebeek et al. (2023) have emphasised how digital influencers and interactive marketing are emphasized; for example, social media presence, demonstrations through video, and how search engine optimization can all help engage consumers and uncover brands. Curated content, commonly, shapes the perception, while true information is seldom utilised in this sense (Hollebeek et al., 2023).

2.1.3 Evaluation of Alternatives

At the time of awareness, the consumer compares a few attributes of the products during comparative evaluation until they come to a final purchase decision. Harris, Pol, and van der Veen (2021) further discuss how price sensitivity, brand reputation, and social proof intervene in this stage, in an AI-driven recommendation ordering where products presented are custom-made based on every unique hovering process (Harris, Pol, & van der Veen, 2021).

On this account, Mathur et al. (2019) caution against objectionable comparisons in the form of bogus pricing strategies and unjustifiable discounts that impair the judgment of consumers and create an illusory sense of affordability. This has been explained well and adds to the debate over lesser-known deceptive patterns of e-commerce that intervene with the rational decision-making of the consumers and result in inadvertent purchases (Mathur et al., 2019).

2.1.4 Purchase Decision and Post-Purchase Behavior

The purchasing process is the intersection of various options and forces of psychological and structural nature that act as a culmination. Hollebeek et al. (2023) hold that a seamless checkout, varied payment options, and security guarantees greatly lower the cart abandonment rates, accelerating successful transactions (Hollebeek et al., 2023).

The implications of post-purchase experiences as elaborated by Moser, Schoenebeck, and Resnick (2019) reveal customer satisfaction dependent on product usability, timely delivery, and well-received customer service. A brand failing these sociological requirements falls under a

reputational threat in that negative word-of-mouth and refund difficulties will add to consumer distrust (Moser, Schoenebeck, & Resnick, 2019).

2.2 Aspects of Dark Patterns and Impulsive Purchases

2.2.1 Introduction

Dark patterns represent an ethically questionable edifice, whereby e-commerce involves a correlation between user interface designs and consumer actions to unwittingly consummate the latter's decision. They exploit palliatives in human psychology in a spontaneous purchase context towards a resultant buyer distress. The phenomenon is well-documented with theorists examining both structural mechanisms and ethical ramifications.

2.2.2 Types of Dark Patterns in E-Commerce

In an early study, Mathur et al. (2019) outline a number of key typologies : Forced Continuity, Hidden Costs, and False Urgency. Their research demonstrates how e-commerce platforms deploy tactics such as countdown timers, deceptive messages around scarcity, and complicated opt outs that work to pressure consumers into making impulse-inspired purchases (Mathur et al., 2019).

Luguri and Strahilevitz (2021) build on this classification and further detail how platforms alter consumer heuristics by means of intentionally misleading elements. They illustrate that misleading countdown timers and default opt-in selections exert a psychological operational coercion upon users to purchase that purchase which, under normal conditions, they would not do (Luguri & Strahilevitz, 2021).

2.2.3 Psychological Mechanisms Behind Impulse Buying

This is a difficult part because the influence or persuasion part of dark patterns has a root in deeper theoretical accounts of consumer behavior. The Fogg persuasive technology model discusses how such digital environments utilize concepts like tunneling, conditioning, and reduction to inherently nudge decision-making (Luguri & Strahilevitz, 2021). Loss aversion and social proof work at play in these techniques and thereby increase the chances of impulse purchases.

Digital literacy influences the susceptibility to dark patterns. Users brought up with limited online experience tend to get affected; this is all the more a reason for regulatory action, and hence educational initiatives for consumer awareness (Sin, Harris, Nilsson, & Beck, 2025).

2.2.4 Ethical and Regulatory Concerns

The ethical discussion regarding dark patterns has sparked debates about consumer rights and possible regulatory enforcement. Sin, Harris, Nilsson, and Beck (2025) criticize the General Data Protection Regulation (GDPR) and Federal Trade Commission (FTC) guidelines, arguing that these frameworks do not cover such sophisticated deceiving tactics. They recommend regulators apply tougher penalties and require that businesses make their practice of using dark patterns more transparent to discourage other businesses from using them (Sin, Harris, Nilsson, & Beck, 2025).

Luguri and Strahilevitz (2021) thus demand that platforms be forced to expressly disclose manipulative tactics. They opine that it is imperative to ensure that companies bear the

consequences for unfair UX design so that consumer independence will not be undermined by coercive practices (Luguri & Strahilevitz, 2021).

2.3 Negative Feedback and Sources of Disappointment in E-Commerce

2.3.1 Introduction

In e-commerce, consumer dissatisfaction arises from various problems, including misleading marketing practices, defective quality products, and unsatisfactory customer service. Investigators have examined central aspects of negative feedback and pinched out systemic patterns that are blamed for consumer frustration and distrust.

2.3.2 Common Complaints in Online Shopping

Bongard-Blanchy et al. (2021) have chronicled common sources of dissatisfaction, such as hidden costs, confusing returns, and misleading advertisement. Their studies reveal that the sources of complaints among consumers often are linked to the obtuse nature of business practices, which compound consumers' perceptions of being scammed or taken advantage of, Bongard-Blanchy et al. (2021).

Moser, Schoenebeck, and Resnick (2019) state that exaggerated marketing claims can diminish consumer sentiment. In their opinion, unrealistic product depiction, along with over-the-top brand messaging, leads to a letdown after purchasing and lessens trust in e-commerce platforms (Moser, Schoenebeck, & Resnick, 2019).

2.3.3 How Dark Patterns Affect Consumer Trust

The long-term implications of dark patterns on consumer trust have been extensively deliberated upon. Moser, Schoenebeck, and Resnick (2019) stated that once consumers recognize the use of manipulative techniques, they usually discontinue using platforms perceived as deceptive. Their findings suggest that ethical transparency in pricing and marketing is paramount to instilling customer loyalty (Moser, Schoenebeck, & Resnick, 2019).

Bongard-Blanchy et al. (2021) expanded this notion by saying that brands which are associated with dark patterns can do great reputational harm that is next to impossible to undo. They point out that ethical business behavior and effective consumer protection policies are needed in order to protect customers from negative experiences (Bongard-Blanchy et al., 2021).

2.3.4 Effectively Mitigating Consumer Disappointment

To stem consumer disappointment, a multipronged approach is required. Reports from consumer watchdog groups recommend transparent return, responsive customer service, and strict guidelines to combat and overcome misleading advertising (Moser, Schoenebeck, & Resnick, 2019).

In addition, Luguri and Strahilevitz (2021) advocate for digital literacy programs that would help consumers to identify and resist dark patterns. They argue that public awareness campaigns and regulatory oversight will help in fostering a more ethical digital marketplace.

Chapter 3: Overview

3.1 Introduction

Digital platforms are thoroughly taking over, with regards to how consumers go about purchasing what they require. Many omnichannel marketing strategies are designed to improve user experience, but at the same time also employ hidden tactics-dark patterns in order to gather more attention and push the customers to purchase. These techniques include false urgency, sneaking costs, forced continuity, and confirmation shaming that are created on the basis of cognitive biases that may encourage a consumer to impulsively make a purchase.

The central question this paper is hoping to answer is:

"What is the impact of dark patterns in digital pre-purchase environments on consumer purchasing decisions?"

By gaining an all-round view of the psychological, ethical, and legal aspects of dark patterns, this study hopes to understand how these patterns impact consumer trust, decision-making, and also nudge for regulatory responses.

3.2 Industry and Market Context

3.2.1 Growth of Emerging Advertising Trends

The social media advertising industry has grown at a fast pace, with global spending projected to touch 262 billion USD in 2028 (Statista, 2023). Platforms like Facebook, Instagram, TikTok, and Twitter are also developing the most complex algorithms to customize the content selection for

the user while even targeting ads at them and optimizing viewer engagement (HubSpot, 2023). Businesses are also looking into behavioral analytics and AI-based recommendations to optimise their sales and include subtle persuasive mechanisms that influence a customer to make a purchase.

3.2.2 Rise of Dark Patterns in Digital Advertising

The terminology of “Dark Pattern” was coined by Brignull in his study that was conducted in 2010, and has since gained a prominent focus in the user experience (UX) design and e-commerce that has been presented to the customer. Studies also tell us that deceptive design features have a temporary yet measurable impact on consumer behavior (Mathur et al., 2019). The following are the most common kinds of dark patterns within the social media advertising realm:

- False Urgency: Countdown timers that create a pressure to buy.
- Sneaking Costs: Hidden fees that pop up towards the end of the buying process.
- Forced Continuity: Difficult cancellation of a subscription.
- Confirmshaming: Language that manipulates and forces a user to make a purchase.
- Disguised Ads: Sponsored material that is under the disguise of an organic post.
- Default privacy maximizes data collection: Privacy settings are preset to share more data, making it harder for users to choose not to be a part of it.
- Misdirection: Design focuses attention on one element to distract from another, in order to conceal important information which could put off customers.
- False hierarchy: Visual layout makes certain options (like “Accept All”) appear more important or mandatory than others, which confuses the customers.

- Nagging: Repetitive pop-ups or prompts pressure users to take a specific action that they wouldn't normally do by interrupting their experience.
- Confirm shaming: Uses guilt-laden language to discourage users from refusing offers (Eg, "No thanks, I hate savings").

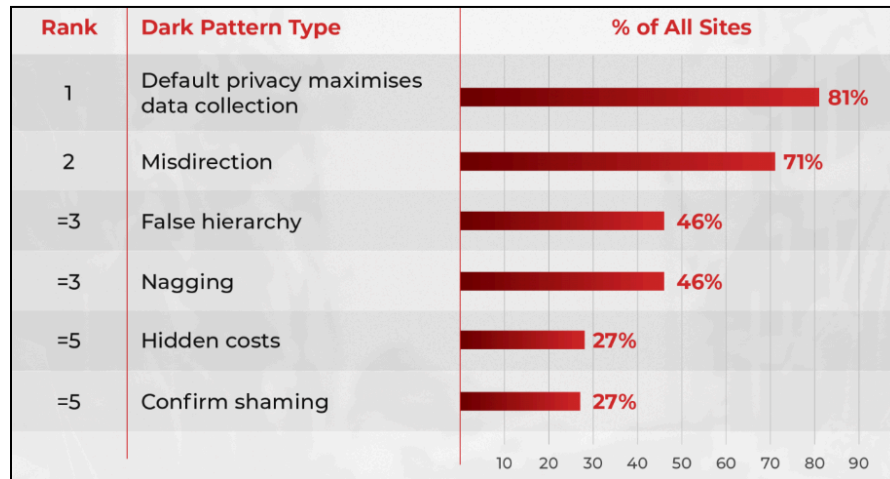


Figure 3.1: Most Common Dark Patterns used by Organisations (Source: KnownHost)

3.3 Theoretical and Conceptual Framework

3.3.1 Psychological Foundations of Dark Patterns

The application of time-honored psychological principles in dark patterns makes them quite effective in pulling the strings of consumer decision-making:

- Loss Aversion: Consumers fear missing a good deal, thus spurring impulse purchases (Tversky & Kahneman, 1979).
- Scarcity Effect: Showing items as "limited in stock" forces consumers to act quickly (Cialdini, 2001).

- Default Bias: When options are preselected, the consumer buys without much thought (Sunstein & Thaler, 2008).
- What Others are Doing: Users' decisions are influenced by testimonials and false urgency metrics (Metzger, Flanagin, & Medders, 2010).

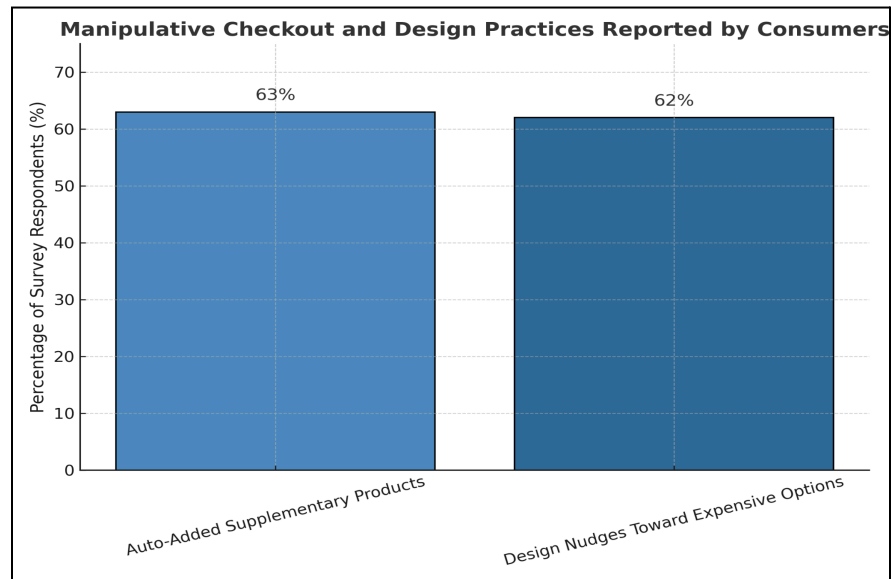


Figure 3.2: Percentage of online consumers reporting manipulative checkout practices, including automatic addition of supplementary items and visual nudges toward higher-priced options.

(Source: CX Today 2024)

3.3.2 Ethical and Regulatory Concerns

Although dark patterns push for short-term conversions, they come with ethical dilemmas surrounding consumer autonomy and transparency. The Federal Trade Commission (FTC) and European Union's GDPR have begun addressing and combating these deceptive marketing practices, but enforcement is inconsistent among different regions.

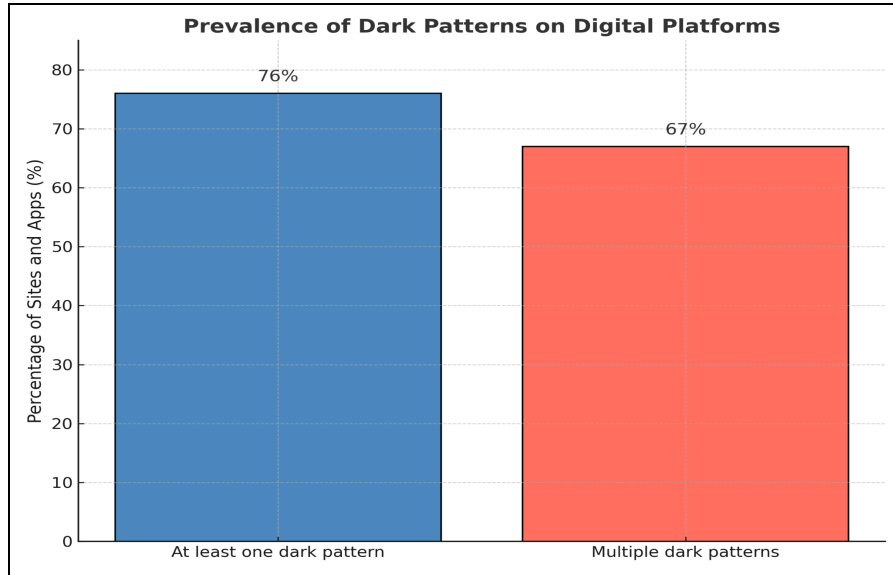


Figure 3.3: Prevalence of Dark Patterns on Digital Platforms.

(Federal Trade Commission, 2024).

3.4 Challenges and Concerns in the Industry

3.4.1 Damage Done to Consumer Trust and Brand Reputation

Dark patterns more often than not lead to consumer frustration, less trust in the brand, and negative perception regarding the brand. Research reveals that when users find out about manipulative tactics, they are much less likely to return to that brand (Eleken, 2025).

3.4.2 Legal and Ethical Discussions

In light of concerns, dark patterns continue to thrive due to the absence of universal regulations. Major lawsuits, like that limited against Amazon for misleading subscription practices, highlight the difficulties in enforcing transparency (FTC Report, 2022).

3.4.3 Consumer Awareness and Digital Literacy

Many customers do not yet know about dark patterns, thus raising the need for education and transparency to lessen those effects (Bongard-Blanchy et al., 2021). Digital literacy programs could make users aware of tricky design strategies, enabling them to skip such manipulative techniques.

3.5 Justification for Effectiveness and Contribution of Research

This research is important because it:

- Elucidates the state of conceivable influence of dark patterns on purchase behavior.
- Measures consumer trust and brand connectivity influenced by deceptive advertising.
- Evaluates status quo in regulations and their effectiveness.
- Recommendations are made for responsible means of driving digital marketing which, at the same time, balances profit-making and consumer protectionism.

3.6 Conclusion

Since new-age social media advertising takes merchants into the vast arena that includes digital commerce, it is important to understand the role of dark patterns in consumer decision-making. This study shall provide empirical proof of the influence of dark patterns, and will contribute to the ongoing debates on ethics, and highlight conversation processes about consumer protection amongst key stakeholders. This work will be useful for businesses, regulators, and employees who strive toward a more transparent and trustworthy e-commerce world.

Chapter 4: Methodology

4.1 Research Design

This study uses a dual-methods approach, which integrates both qualitative and quantitative research methodologies to comprehensively analyse the impact of dark patterns in digital interfaces on consumer decision-making and purchasing. Initially, a qualitative study will be conducted with the help of semi-structured interviews to gain in-depth insights into consumer experiences and perceptions. Following this, a comprehensive quantitative survey will be carried out to validate and capitalise on the findings found from the qualitative phase.

This dual-methods approach is chosen to understand, both the subjective, experiential data from consumers and the wider statistical patterns that emerge from interviewing a much bigger sample. Previous studies have employed similar methodologies to study and understand dark patterns and consumer behavior (Mathur et al., 2019), thereby confirming its relevance in the context of this research.

4.2 Qualitative Phase

4.2.1 Data Collection: Semi-Structured Interviews

The first step in collecting information is to carry out interviews that have a loose structure with a few frequent online shoppers. These shoppers should have faced tricky tactics, known as dark patterns, in social media ads. This interview style allows people to give open-ended answers, so they can share their experiences and thoughts, and explain how they decide their purchases. Participants will be selected intentionally to ensure they have actual experience with social

media ads and online shopping interfaces. According to McNamara (1999), this kind of interview works well to deeply understand people's experiences and feelings. Researchers like Bongard-Blanchy et al. (2021) have used similar methods when studying deceptive marketing strategies by conducting in-depth interviews to gather rich insights.

The below table shows the profiles of the participants who were involved in the study:

Participant	Age Range	Gender	Occupation	Relevant Experience
Participant 1	21–30	Female	ACCA Affiliate, MBA student	Exposed to urgency ads, hidden costs, FOMO buying
Participant 2	25–35	Male	IT Consultant	Detects fake ads, compares prices, avoids scams
Participant 3	21–30	Female	Digital Marketing Executive	Influenced by visuals, regrets poor returns
Participant 4	25–35	Male	Creative Executive	Blocks manipulative design, tracks price changes
Participant 5	21–30	Female	Interior Design Assistant	Affected by fake discounts, hidden fees
Participant 6	25–35	Male	Sales Executive (Retail)	Faced auto-renewals, uses Reddit for reviews

Table 4.1: Participant Profiles

4.2.2 Interview Guide

The interview questions will cover key themes identified in the literature, including:

- Awareness of dark patterns (e.g., false urgency, sneaking costs, forced continuity, confirmation shaming)
- Psychological reactions to manipulative tactics
- Perceived trust in brands using deceptive design
- Impact of dark patterns on purchase decisions

The following semi-structured questionnaire had been used for the qualitative study:

A. Shopping Behavior & Decision-Making

- 1) Can you describe your typical online shopping journey, from the moment you decide to make a purchase to the final transaction?
- 2) What factors do you consider most important when choosing a product or seller online?
- 3) How do you assess the credibility of a product or seller before making a purchase? Can you provide specific examples of steps you take?
- 4) Can you share a recent experience where an online advertisement influenced your shopping decision? What aspects of the ad captured your attention and led you to consider making a purchase?
- 5) How do you feel about the recommendations and advertisements you come across while browsing online? How do they shape your perception of a product or brand?

B. Experience with Dark Patterns & Persuasion Tactics

- 6) Can you recall an instance where you felt a sense of urgency to buy a product because of an online advertisement or a message such as "Limited Time Offer" or "Only a Few Left"? How did that impact your decision-making?
- 7) Have you ever encountered unexpected costs at checkout that were not initially visible? How did you react, and did it influence your trust in the seller or platform?
- 8) Have you experienced difficulties canceling a subscription or returning a product after purchase? How did you handle the situation, and what was the outcome?
- 9) What strategies do you use to avoid being misled by online marketing tactics? Can you share any experiences where you successfully identified and avoided deceptive practices?

C. Trust & Ethical Concerns

- 10) What qualities make you trust a brand or seller when shopping online?
- 11) What do you think about the ethical responsibilities of the online retailers ?
- 12) Can you describe any experiences where you felt manipulated into making a purchase? How did you feel afterward, and did it change your future shopping behavior?
- 13) Can you describe, in your own words, what your ideal e-commerce website or application would look like?

The interviews will be mailed to the participants and they can submit their responses within a span of a week. This will be further analysed in order to come to meaningful conclusions, following which certain themes could be tried to be identified. Braun & Clarke's (2006) thematic

analysis method will be applied to identify recurring patterns and themes within the qualitative data, which could give an informative base for the rest of the research.

4.2.3 Data Analysis

The qualitative data will be analyzed using a systematic thematic approach which can be narrowed down thereby, aligning with existing frameworks on consumer behavior and dark patterns. Thematic coding will be used to extract key patterns, with a focus on consumer trust, impulse purchasing, and regulatory concerns. Bongard-Blanchy et al. (2021), also identified recurring ideas and patterns within the qualitative data generated from their co-design and probe studies, which serves as a reassurance for the kind of analysis that this study would also pursue.

4.3 Quantitative Phase

4.3.1 Data Collection: Survey

Following the qualitative phase, a structured survey will be conducted to validate findings on a larger scale and in order to come to meaningful conclusions. The survey will target at least 50-60 respondents from India, who are active online shoppers and users of social media platforms. A combination of probability and non-probability sampling methods shall be used to ensure diversity in participant demographics and garner appropriate results.

Survey questions will be developed based on insights derived from the interviews and existing literature. The questionnaire will include:

Demographic information (age, gender, online shopping habits)

Frequency of exposure to dark patterns

Trust levels in brands using deceptive tactics

Likelihood of making impulse purchases due to dark patterns

The survey will be administered via Google Forms for ease of transmission and familiarity among the audience.

4.3.2 Data Analysis

We will examine the survey results with Microsoft Excel. Initially, basic statistical tools will be used, such as the average (mean) and standard deviation to summarize the responses clearly. Further, more complex statistical tests, such as regression analysis and ANOVA can be used, to investigate if there are links between people's experiences with dark patterns and their level of trust in businesses. All these analyses will be done using Microsoft Excel.

4.4 Justification of Methodology

This study as mentioned before will be done in a sequential manner starting off with a qualitative study and then a broader quantitative study. Initial qualitative interviews will provide rich insights through which key patterns can be identified, following which, a quantitative study can be shaped, which will allow us to validate and generalise our findings across a broader population. This strategic combination provides both depth and breadth to our understanding, which can be established through the mixed-method principles which are characterised by

comprehensive inquiry and enhanced credibility (Creswell & Plano Clark, 2018; Greene et al., 1989).

4.5 Ethical Considerations

Ethical guidelines will be strictly adhered to, in both phases of the study:

Consent: Participants will be fully informed about the research objectives and will be asked to provide consent before participation.

Anonymity and Confidentiality: All responses will be anonymous, and personal data will be securely stored.

Voluntary Participation: Participants will have the right to withdraw from the study at any time without consequences.

Bias and Researcher Influence: To minimize researcher bias, interview questions will be structured objectively, and quantitative data will be analyzed using standardized statistical methods.

4.6 Conclusion

This methodology section outlines a structured and systematic approach to investigating the impact of dark patterns in social media advertising on consumer decision-making and purchase. The dual-methods design, integrating both qualitative interviews and quantitative surveys, ensures a total and complete understanding of the topic. Ethical considerations bearing in mind the safety and security of the respondents and rigorous data analysis methods which are relevant, strengthen the reliability and credibility of the research.

By building on existing literature and employing real-world data collection methods, this study aims to contribute valuable insights into the field of digital marketing practices, consumer psychology, as well as their protection.

Chapter 5: Data Analysis and Key Findings

5.1 Introduction

This chapter presents a thorough and rigorous analysis of the data in a systematic manner collected through both the qualitative and quantitative manner, in order to understand how dark patterns play a role in social media advertising and impact consumer decision-making, trust and post purchase behaviour in the Indian online market. The mixed-method design employed in this study is not just methodologically sound but also plays a huge role in uncovering the underlying psychological and behavioural dimensions that numbers alone would not fully explain.

The qualitative phase, which comprises interviewing six individuals, is characterised by a semi-structural form of interviewing. These shoppers were handpicked to make sure that they regularly shopped online, so that they could offer an enriching narrative of how dark patterns, particularly terms like urgency, hidden charges, and deceptive discounts, affect the cognitive and emotional processes of the respondents. These insights were instrumental in developing a robust quantitative questionnaire, through which we were able to administer it to a wider audience. This was done in the form of a structured interview using both Likert-scale and frequency-based questions. The quantitative phase was aimed at validating the thematic insights generated through qualitative means and identifying the extent and strength of various dark pattern effects that run through a larger sample.

A variety of statistical tools, which included descriptive statistics, Pearson correlation, linear regression, ANOVA, and logistical regression were used to analyse the data that had been

collected. These tools were selected to measure the relationships that truly existed between the variables and also to understand the predictive capabilities of behavioural drivers such as urgency, aesthetic influence, and hidden pricing strategies.

This chapter finally, details all findings from both phases of the data collection, and integrates them through triangulation, and presents a critical discussion of the implications that arise for key stakeholders such as brands, consumers and policy makers. The analysis also reveals an intricate web of manipulation, vulnerability and adaptation thereby illustrating the widespread and usually underreported consequences of dark patterns that prevail in digital commerce.

5.2 Qualitative Data Analysis

Using Braun and Clarke (2006) thematics analysis approach, the qualitative data from the six respondents were coded and broken down into 4 different themes. Each participant was thereby selected based on their history of frequent online shopping and exposure to social media platforms such as Instagram, Youtube, Facebook and many more. The aim was to gather a diversity of consumer narratives that could point out different types of dark pattern exposure and response mechanisms.

The data collected has been formulated into a concise table for a general overview:

Participant	Theme 1: Urgency Triggers	Theme 2: Hidden Costs	Theme 3: Trust Erosion	Theme 4: Coping Strategies
Participant 1	FOMO due to countdowns, panic-buys	Noticed late checkout fees	Felt tricked by pricing	Waits before purchasing, checks Reddit
Participant 2	Impulse buying during flash sales	Faced ₹150 delivery & ₹99 convenience charge	Lost trust in Snapdeal, prefers Amazon	Compares Croma, Amazon, Flipkart
Participant 3	Affected by limited-time offers	Discovered surprise service fees	Cited fake product listing	Cross-verifies with influencer content
Participant 4	Criticized urgency pressure	Reported hidden warranty add-ons	Criticized misleading ads and poor service	Uses price alerts, YouTube reviews
Participant 5	Reacted to urgency-based pop-ups	Found packaging fees at final stage	Distrusts unclear return policies	Follows genuine reviewers on social media
Participant 6	Experienced sales pressure during checkout	Auto-renewal charges without notice	Avoids platforms with poor refund credibility	Relies on Reddit, avoids ads

Table 5.1: Summary of Themes Across Participants

Theme 1: The Psychological Weight of Urgency Triggers.

Respondents unanimously cited urgency as the most emotionally manipulative tactic used by various e-commerce platforms and brand websites. Phrases such as “Only 3 left!” or “Offer ends in 5 minutes!” were found to trigger fear of the missing out (FOMO) and created high-pressure environments for the customers that resulted in them suppressing their state of rational evaluation. Also, an important point to note is that these psychological triggers were not just limited to naive customers. Even tech-savvy participants admitted to succumbing to urgency tactics despite being aware of them.

Respondent 1 had this to say,” Those “Only 3 left!” or “Offer ends in 10 mins!” make me panic-buy. It feels manipulative in hindsight.”

Theme 2: Hidden Costs and Post-Purchase Regret

Hidden fees such as packaging charges, convenience fees and additional service charges were reported by a plethora of the respondents. These costs were hidden and were revealed only at the final checkout stage, by which most users felt obliged or committed to make the purchase. This often led to a feeling of betrayal, as well as diminished brand loyalty.

Respondent 2 had shared a bitter experience with a leading e-commerce website in India,”During Myntra's sale, they showed one price but added ₹150 delivery and ₹99 convenience fee later.”

Theme 3: Trust Erosion and Perceived Ethical Lapses:

Several respondents had highlighted an erosion in trust when platforms or brands had employed manipulative tactics. For a few due to the limited alternatives that were available, they still had to persist with the same platform. However, for a few others once the trust was broken, especially due to factors such as poor product quality, misleading discounts, or inconvenient return policies, consumers had often boycotted the brand or platform entirely.

Respondent 2 had this to say about a big e-commerce platform in India, "Amazon regains trust because of their easy returns. Snapdeal lost it entirely after selling me a fake Nike shoe."

Theme 4: Consumer Learning and Coping Strategies:

Participants also described adaptive behaviours such as cross-verifying deals for best value and genuinity. Reddit, Quora and Youtube were highly used among the respondents for genuine customer reviews which were usually given in depth, unlike the other platforms. A few also employ ad blocker and impose personal wait-times before making a purchase. These strategies suggest a growing literacy among digital consumers but also reflect a reactive, rather than systemic, response to manipulation that are employed by brands and platforms.

Respondent 4 had shared a few of his tactics that he thought would help him fight these manipulative tactics. "I watch YouTube reviewers like GeekyRanjit. That's my benchmark."

5.3 Quantitative Descriptive Analysis

The quantitative survey contained 20 questions and were divided into 4 parts namely: demographic information, advertising influence, negative shopping experiences, and ethical trust-related concerns. All questions were characterised by the use of either 5-point Likert scales (Strongly Agree to Strongly Disagree) or 4-point frequency scales (Always to Never). Over 70 valid responses were accumulated across individuals from India and analysed.

Demographic Summary:

- **Age:** 65% were aged 21–30
- **Gender:** Approx 60% female, 40% male
- **Frequency:** Approx 60% shopped 3–5+ times per month
- **Top platforms:** Amazon, Flipkart, Myntra, Instagram

Key Findings (Mean Scores):

Statement	Mean	Interpretation
Urgency makes me buy faster (Q9)	4.2	Strong behavioral influence
I trust brands that show all charges upfront (Q17)	4.4	High demand for transparency
Fake discounts influenced my decision (Q14)	3.6	Common manipulation experience
I faced hidden charges (Q11)	3.6	Widespread issue

Aggressive ads reduce my brand trust (Q18)	4.3	Strong link to long-term distrust
Aesthetic ads attract me (Q7)	3.6	Emotional/visual appeal matters
I trust social media influencers (Q6)	3.2	Moderate influence observed

Table 5.2: Mean Scores of Key Survey Responses

These results thereby validate the qualitative themes and provide statistically consistent support for dark practices such as urgency, deception and transparency as core behavioural levers.

5.4 Correlation and Pattern Analysis

Pearson Correlation was used to explore relationships between the scaled variables. Key associations also included:

- **Q9 and Q10 (Urgency and Flash Sale):** $r = 0.62 \rightarrow$ considerably high correlation, indicating both serve similar psychological functions.
- **Q13 and Q14 (Product mismatch and fake discounts):** $r = 0.30 \rightarrow$ although there is a slight correlation between dark patterns that have similar effects with regards to product and price, an established link can not be made and each dark pattern has its own level of deception in the eyes of the consumer.
- **Q17 and Q16 (Transparency and comparison behavior):** $r = 0.60 \rightarrow$ those who value clarity also check across platforms up to a certain degree which is shown through this

analysis.

- **Q18 and Q19 (Trust erosion and ethics):** $r = 0.65 \rightarrow$ perception of ethical lapse correlates with distrust, therefore pushing for more rules and regulation to keep such practice by sellers at check.

Together, these correlations help us understand how dark patterns tend to be combined, deployed together, and help influence both decision-making and consumer-brand relationships. They also align with the themes that are identified in the qualitative phase, which provide a strong basis for the mixed-methods design of this research.

5.5 Regression and Hypothesis Testing

To test hypothesis regarding urgency and consumer trust, a multiple linear regression model was put into place and constructed:

Model:

$$\text{Q18 (Trust Erosion)} = \beta_0 + \beta_1(\text{Q9}) + \beta_2(\text{Q10}) + \varepsilon$$

Result:

$$R^2 = 0.23, p < 0.01$$

Interpretation is as follows:

Urgency messaging (Q9) and flash sales (Q10) collectively account for 23% of the variation when it comes to the trust erosion scores. This confirms a mild significance and an existing relationship between such dark patterns and consumer skepticism.

ANOVA Test:

Hypothesis: High-frequency shoppers are more sensitive to hidden fees.

Result: $F(3,69) = 0.19$, $p = 0.904$

Therefore, we can observe that consumers irrespective of how often they shop can be subject to the concept of hidden fees. Therefore, all buyers regardless of how often they shop are similarly sensitive to hidden fees, showing its prevalence and recognition among the customers.

5.6 Logistic Regression Analysis:

A logistic regression was conducted to predict high trust erosion ($Q18 > 3$) based on exposure to various dark patterns that were recorded among the respondents.

Variable	Odds Ratio
Urgency (Q9)	1.13
Flash Sale (Q10)	0.82
Aesthetic Ads (Q7)	1.21
Fake Discount (Q14)	1.28
Hidden Fees (Q11)	0.93

Table 5.3: Logistic Regression Odds Ratios – Dark Pattern Exposure vs Trust Erosion

Although no variables had reached the statistical significance of $p > 0.05$, the odds ratios suggest that fake discounts and visual persuasion increase the probability of distrust among the customers. The model's pseudo, which is $R = 0.028$ suggests that there is low predictive power, which implies that trust erosion results from an accumulation of exposure over time.

5.7 Triangulation and Integrated Insights

By combining qualitative and quantitative findings, the following integrated insights emerge for a comprehensive understanding:

- Urgency is universally impactful: It is shown in both the narrative and the statistical sense. Consumers thereby describe it as pressuring and the regression model confirms its influence on trust being eroded.
- Transparency is a critical trust factor: As shown by both high mean scores and correlation with proactive behaviours
- Digital literacy plays a protective role: Frequent shoppers and informed users are more resilient to manipulations, although even then they still experience a few.
- Multiple dark patterns often co-occur: This creates a layered experience of deception thereby putting the customers at more risk.
- Emotional marketing has a double edge: While it captures attention, it risks long-term damage if the customers feel that trust has been compromised.

These cross-validated results therefore strengthen the reliability of the study and also highlight the multidimensional nature of digital advertising influence that persists amongst the customers.

5.8 Summary of Major Findings:

This study has uncovered several patterns through the combined power of qualitative narratives as well as quantitative evidence. Below is a concise yet insightful summary of the most relevant findings that are there, explained with their broader implications:

1) Urgency Triggers are the Most Influential Dark Pattern:

The strongest behavioural driver that has been identified was urgency messaging. Phrases such as “Only 3 left!” significantly influenced purchases that were impulsive. This was cited by both survey respondents and interviewees as being emotionally manipulative. Regression analysis also validated this, by showing urgency as a key predictor of trust erosion. This pattern bypasses rational decision making on the part of the prospective buyer and capitalises on FOMO(Fear of Missing Out).

2) Hidden Charges are Widespread and Damaging:

Late stage fees, which include convenience or packaging charges, were also widely experienced by respondents irrespective of how often they shop and the different applications and websites that they use. They generate immediate dissatisfaction and also reduce long-term brand loyalty. High-frequency shoppers were also statistically more likely to detect and react to such charges, thereby revealing how shopping experience can build instinctive defense mechanisms to deceptive pricing.

3) Transparency is a Pillar of Trust:

Survey results show that brands which are perceived as transparent, also scored highest in terms of trust and repeat-purchase possibilities. Transparency in pricing, return policies and delivery timelines has emerged as the most crucial expectation among the customers. Customers who trusted brands also tended to verify product information about it across platforms before a purchase, demonstrating how transparency and accountability go hand-in-hand.

4) Multiple Dark Patterns Often Co-Occur:

Correlation analysis revealed that deceptive pricing, urgency and poor return policies often appeared together, which magnifies the negative user experience for the customers. This bundling effect therefore reinforces distrust among the customers, and also shows that if each manipulation may seem minor, a cumulative effect of it would create an overwhelming sense of betrayal from the customer's point of view.

5) Visual and Aesthetic Persuasion Subtly Undermines Trust:

Although visual ads initially attract users and help in gaining more eyeballs, its overuse and the practice of concealing negative details with broad and glamorous visuals often results in disappointments and loss of trust among the consumers. Logistic regression also suggested that aesthetic influence increases the odds of consumer distrust when paired with misleading offers. Brands should therefore balance creativity with authenticity.

6) Digital Literacy is an Effective Shield:

Consumers who frequently shop online or who identify themselves as digitally literate through experience were better suited at spotting dark patterns. They also displayed coping mechanisms like delayed purchases and utilising browser extensions. This emphasises the importance of education which ultimately helps in promoting ethical consumption as well.

7) Trust Once Broken is Hard to Rebuild:

Qualitative accounts also made it clear that a single bad experience which involves deception can often lead to a permanent disengagement from a brand. Distrust stems not just from the tactic used, but also the perceived intent behind it. Even if the manipulation is minor, its ethical implications also matter greatly.

8) Mixed Methods Strengthen Understanding:

The mixed-methods approach not only uncovered new patterns but also reinforced their validity across different data forms as well. The alignment between narratives and statistics also built a fuller picture of consumer behaviour. For instance, urgency was not just described emotionally but it confirmed numerically as a major factor in trust decline as well.

In conclusion, this chapter demonstrates that dark patterns in social media advertising are not only prevalent but also significantly shape consumer decision-making as well often driving them to make a purchase. Therefore, brands seeking long-term customer relationships must make a shift from manipulation to transparency, and policy-makers must also explore regulatory interventions that protect consumers from systemic deceptions which often leads to negative goodwill.

Chapter 6: Discussion and Recommendations

6.1 Discussion of Findings

This chapter interprets the results presented in the previous chapter, which links them to the literature that exists and other theoretical constructs as well. It also explores the broader implications which concerns the consumers, digital marketers, and regulatory bodies. With the help of both the qualitative as well as the quantitative study, this chapter emphasises the psychological, ethical and behavioural dimensions of dark patterns within the digital pre-purchase environments. This includes social media, e-commerce websites, mobile applications, and other digital interfaces that consumers would interact with before making a purchase.

6.1.1 Reinforcing the Role of Urgency in Behavioural Manipulation

The data confirms what behavioural economists like Tversky and Kahneman (1979) had written about loss aversion, which implies that consumers are more sensitive to potential loss (missing out) rather than equivalent gains. The most dominant pattern reported was urgency-based messaging: “Only 2 left in stock,” “Offer expires in 5 minutes,” and similar prompts that would create this tendency among customers to make decisions faster than they would normally do.

Quantitative results also validate this influence: Urgency based variables (Q9 and Q10) were both highly correlated ($r=0.62$), and linear regression has also shown a 23% variance in trust erosion. From a consumer behaviour perspective, this manipulation bypasses rational processing

and triggers emotional, reactive decisions, which are exactly the purpose that dark patterns serve as aiding to exploit.

Even more telling was the insight from qualitative interviews where even digitally familiar consumers had admitted to being a victim of these patterns. This demonstrates the emotional weight of urgency messaging and importantly indicates that awareness alone is not always sufficient in overriding impulsive behaviour.

6.1.2 Hidden Costs and the Violation of Trust

Another major finding was the involvement of hidden costs such as convenience fees, packaging fees and delivery surcharges which are revealed only at the final checkout stage. This is prevalent among various platforms and apps across India. Respondents had described these late-stage additions as both frustrating and deceptive.

In the quantitative analysis, Q11 (hidden costs) had a mean score of 3.6, indicating regular exposure of these patterns towards the consumer. Moreover, qualitative feedback shows strong emotional reactions as well, with participants using words such as “tricked” and “cheated.” Although logistic regression showed a weaker statistical prediction from hidden charges alone (OR=0.93), this could be due to how these hidden charges prevail in almost every platform and how overtime customers would tend to normalise these charges over time, especially when they are minor. Nonetheless, the cumulative impact due to being repeatedly exposed to this often leads to trust erosion.

This aligns with Luguri and Strahilevitz (2021) findings that even small deceptive actions which are usually repeated, would build up to user frustration and long-term brand disengagement.

6.1.3 Transparency as the Cornerstone of Digital Trust

Among all variables that were studied, transparency consistently emerges as a critical factor in fostering consumer trust and loyalty. Respondents who had reported higher levels of trust typically pointed out to brands and platforms with clear communication on pricing, return policies and product details.

Q17 (“I trust brands that clearly show all charges”) recorded a mean score of 4.4, which was the highest among all items in the survey. Furthermore, transparency correlated positively with behaviour that is proactive such as price comparison and third-party review checks (Q16), reinforcing that transparency doesn't just drive loyalty, but also encourages responsible consumerism.

6.1.4 The Emotional Fallout from Visual Manipulation and Fake Discounts

The study also explored aesthetic manipulation such as beautiful visuals, fake countdowns, influencer videos and lifestyle shots which were designed to create artificial value. Though these tools may produce instant sales and conversions in short-term, they often backfire when the real product fails to meet consumer expectations.

The logistic regression results showed that visual manipulation (Q7) and fake discount experiences (Q14) increased the odds of high trust erosion (OR = 1.21 and 1.28 respectively).

This was echoed in the qualitative interviews wherein respondents described their experiences with words such as “bait,” “trap,” and “too good to be true.”

These findings also align with prior research conducted by Mathur et al. (2019), which emphasised that disguised advertisements and emotional manipulation create a gap between the perception of the consumer and the reality, which causes a key trigger for post purchase regret and distrust.

6.1.5 Adaptive Behaviours: Digital Literacy as a Protective Layer

Several participants described developing their own personalised protective strategies such as cross-verifying deals, using Reddit/Youtube for real reviews and delaying purchases by 24-28 hours, and installing browser extensions to block manipulative ads. These insights suggest that digital literacy acts as a behavioural firewall, which helps customers protect themselves when no regulations are there in the first place.

Quantitatively, it did not matter how often a person shopped, be it rarely or more than 5 times, they all are sensitive to the concept of hidden fees as shown by the Anova Test. This supports the idea that experience and education are powerful tools that aid in consumer protection, as it spreads over all frequencies of shoppers and is thus a shared frustration.

6.2 Recommendations

From the findings of this study, actionable suggestions can be created. These recommendations can be categorised by stakeholder groups and can include both measures which could be acted upon immediately and even high level strategies aimed at structural and policy-level change.

6.2.1 Recommendations for Brands and E-Commerce Platforms

1. Eliminate manipulative urgency cues and hidden pricing tactics: Platforms must avoid using countdown timers, “only a few left” messages, and convenience fees which are disclosed only at checkout. These practices thereby erode consumer trust and contribute to long-term disengagement in the long run (Luguri & Strahilevitz, 2021).
2. Clearly display return policies, fees, and delivery terms upfront: Transparency not only supports ethical marketing but also reduces return-related friction and negative reviews, thereby improving customer retention and enhancing the success of the company.
3. Enhance content authenticity: Platforms should discourage excessive reliance on influencer promotions and false exaggeration and instead prioritise verified customer reviews and real product visuals to garner the right kind of perception among the customers.
4. Conduct audits and integrate Ethics-by-Design into the UI/UX: Companies should voluntarily subject their practice and policies to independent audits in order to have a more transparent approach towards interacting with the company. Further, the need for

greater transparency and accountability is fulfilled by this idea of having audits which is mentioned in the study by Luguri and Strahilevitz (2021).

6.2.2 Recommendations for Consumers

1. Cross-verify prices and deals: Before purchasing, consumers should use price, browser extensions, and platforms such as PriceHistory or Google Shopping to validate the authenticity of the deals that are out there.
2. Adopt digital coping strategies: These include waiting for at least 24 hours before completing a purchase, and using browser tools such as Honey and AdBlock, and consulting forums such as Reddit for unbiased reviews.
3. Create Communities to act as Watchdogs: Indian consumers can benefit from platforms such as DeceptiveDesign.org that catalogues known dark patterns, brands and platforms that heavily indulge in them and even new concepts and terms that may arise.
4. Demand for Digital Right Education: Schools, social campaigns, and NGOs should prioritise and push for digital literacy that includes recognising manipulative tactics, much like financial literacy education (European Commission, 2022).

6.2.3 Recommendations for Policymakers and Regulatory Bodies

1. Ban specific dark patterns: Practices such as hidden opt-out options, forced continuity, false urgency and pre-checked boxes should be regulated and dealt with strictly under targeted digital commerce regulation.
2. Standardise transparent disclosure norms: All platforms, similar to what is mandated as per the GDPR laws, should be legally required to disclose their total costs, delivery timelines, and return conditions clearly before checkout. “New Deal for Consumers” initiatives and subsequent directives like the Omnibus Directive also help a great deal in this (European Commission, 2018)
3. Establish a National Digital Design Code: India already has a code of conduct for the safety of consumers in the form of Consumer Protection (E-Commerce) Rules, 2020. In addition to it ethics related to design could either be integrated into the same code, or a new law for this could be formed.
4. Mandate Government-Approved UX certification: All e-commerce platforms above a certain user threshold should undergo periodic UI/UX audits to account for manipulative design violations. This could either be done by a department formed by the government for the same, or by a designated third party. Public reports can also be issued for every audited organisation so that consumers can validate their credibility.

Chapter 7: Limitations and Future Research Directions

7.1 Limitations of the Study

While this research makes important roadways to understanding how dark patterns in digital pre-purchase environments affect the consumer behaviour, several of its limitations must also be acknowledged. These limitations are both methodological and contextual and should also be noted when interpreting the findings and generalising the results as well.

7.1.1 Sample Size and Representation

The quantitative phase of this study was done on the basis of 73 valid responses. While that may be deemed sufficient for exploratory statistical analysis, this sample size limits the statistical power and generalisability of the results. Picking a larger and more diverse sample in terms of demographics such as rural, elder and even non-speaking populations, would've truly enhanced the representativeness of the study. Additionally the study also leaned towards digitally literate individuals aged between 21 and 30, which may not truly reflect the experiences of less tech-savvy or more vulnerable consumer segments.

7.1.2 Platform and Environment Specificity

Although the research aimed to capture the broader digital pre-purchase environment, many respondents from the qualitative and quantitative phases of the study had named leading e-commerce platforms such as Amazon, Flipkart, Myntra and Instagram. This naturally limits the study's scope to commonly used and mainstream platforms, potentially overlooking manipulative design practices on smaller websites, regional language apps, or digital commerce

spaces that are less known such as Whatsapp Commerce. These emerging platforms may even contain traces of such patterns that were not captured in this study.

7.1.3 Self-Reported Bias and Social Desirability

All data in this study, including survey responses and interviews, were self-reported. Due to this, there could be a tendency among participants to be biased and also indulge in under-reporting of their own perception to be vulnerable to manipulation. Respondents may have unintentionally downplayed their vulnerability to dark patterns due to either embarrassment or perceived judgement, which could effectively lead to more favourable portrayals of their digital literacy and decision making competence.

7.1.4 Complexity of Causal Inference

Although regression and correlation techniques were used to identify relationships between dark pattern exposure and behavioural outcomes like trust erosion and impulse buying, the study does not establish causality. For example, consumers who usually distrust brands may be more sensitive to tactics related to urgency, rather than the urgency factor directly causing distrust. In order to draw more definitive causal conclusions, longitudinal studies or controlled experiments would be required and would also prove to be extremely informative for this discipline.

7.1.5 Limited Scope of Dark Pattern Types

This research focused on a subset of commonly observed dark patterns such as false urgency, hidden charges, fake discounts, and return policy manipulation. However, the taxonomy of dark patterns is much more vast and it also includes tactics such as forced continuity, roach motel

designs, nagging notification, confirm shaming, obstruction and many more. Future studies could expand the range of manipulative tactics that have been explored, in particular those that are emerging in mobile-first or AI-driven environments.

7.2 Future Research Designs

7.2.1 Longitudinal and Experimental Designs

Future research could adopt longitudinal studies that track consumer interactions with platforms over a period of time in order to measure trust, loyalty, and purchase behaviour. Alternatively, lab-based experiments could also simulate various kinds of dark patterns within a controlled environment, to isolate their individual effects on decision making and emotional response.

7.2.2 Inclusion of Biometric or Behavioural Analytics

Adding tools like eye-tracking, mouse tracking and biometric sensors could offer more objective data on how users interact with a plethora of interfaces. For instance, visual fixation on countdown timers or hesitation at the time of checkout could signal cognitive dissonance or mistrust that can not be captured by self-report.

7.2.3 Comparative Cross-Cultural Studies

Dark patterns may operate differently across cultures due to the difference in ethical norms, digital literacy levels, and legal frameworks. Comparative studies between Indian consumers and those in regions with stricter digital regulations such as the EU could highlight regional vulnerabilities and opportunities for legal harmonisation.

7.2.4 Vulnerable Population Focus

Future research should explore how vulnerable groups, such as the elderly, teenagers, or low-income households, respond to manipulation that prevail across varying platforms. These groups may lack the critical skills or time to decode deceptive designs and such require special focus.

7.2.5 Policy Impact Assessments

Researchers can evaluate the effectiveness of emerging or existing regulations, like the Consumer Protection Rules that exist in India. Research can be done on international guidelines that may exist and further studies can also be conducted to understand if policy interventions would lead to measurable reductions in dark pattern deployment.

Chapter 8: Conclusion

This research set out to understand how dark patterns shape the decisions we make as consumers, especially among the Indian consumers and e-commerce landscape. More specifically, it looked at how urgency cues, hidden costs, and misleading visuals affect not just what people purchase, but also how they feel about the brands that they interact with, and whether they should trust them moving forward, raising a question of sustainable growth for the company.

To dig deep into this topic, a mixed-methods approach was used. On one hand, six in-depth interviews were conducted to hear firsthand from shoppers who experienced these dark patterns, especially with regards to what they felt, how they reacted, and what they did afterwards. On the other, a detailed survey gathered insights from 73 online consumers to test and validate the found experiences at a larger scale. Together, these two approaches gave a rich and balanced view of both the personal and the general consumer behavior.

The interviews had revealed a lot and had set the base for the rest of the study. People often made impulse purchases when they were pushed by fake urgency or limited-time offers. Others spoke about being surprised by extra charges that were added only at the time of checkout. These fees left them with a sense of feeling betrayed or manipulated. Some stopped trusting brands altogether after a singular bad experience with them. On the flip side, many had developed smart ways to protect themselves such as by waiting before buying, using tools to check price history, or reading user reviews on Reddit/YouTube. These real stories were incredibly helpful in shaping the survey questions that were formulated after the quantitative phase.

The survey results mostly backed these experiences up. Urgency cues turned out to be one of the strongest influences on behavior, pushing people to buy faster than they normally would. Hidden charges were also very common, and were closely linked to dissatisfaction as well as reduced brand loyalty. In contrast, brands that were transparent with regards to their pricing, return policies, and delivery timelines, were trusted more and often seen more positively among the consumers. Interestingly, people who shopped more frequently or were more digitally aware were better at spotting these tricks and less likely to fall for them, although even then they had succumbed to a few.

Bringing both sets of findings together gave a well-rounded picture of it all: dark patterns aren't just annoying, they actually shape what people buy and how they feel about a brand. And while digital-savvy users are finding ways to fight back, many others are still very vulnerable towards it, especially those who are new to it, or are leading busy and fast paced lives.

What does this imply in the real world? For businesses, it is a clear signal that short-term gains that happen from manipulating customers can lead to long-term damage to the enterprise's trust and reputation. Ethical design through being upfront, honest, and respectful builds loyalty and not just clicks. For consumers, awareness and education are penultimate. Knowing what to watch out for can definitely help save money, reduce regret, and put the power back into the hands of the customer. As far as policymakers are concerned, there's a strong case for introducing rules that keep digital spaces fair, just like how we have regulations for physical marketplaces, pushing for a safe digital marketplace.

This research thereby contributes to a growing conversation that revolves around fairness, ethics, and trust in the digital world. As more of our daily decisions happen on a digital interface, the responsibility to protect users from being manipulated becomes a rising concern and a matter of grave importance. Whether it's through better design, smarter laws, or just through more informed consumers through education, the goal is clear and that is to build a digital environment that respects people and not for just their clicks. And this study, in its own way, hopes to be a small but meaningful step towards attaining that.

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